# Doctor's Helper analysis

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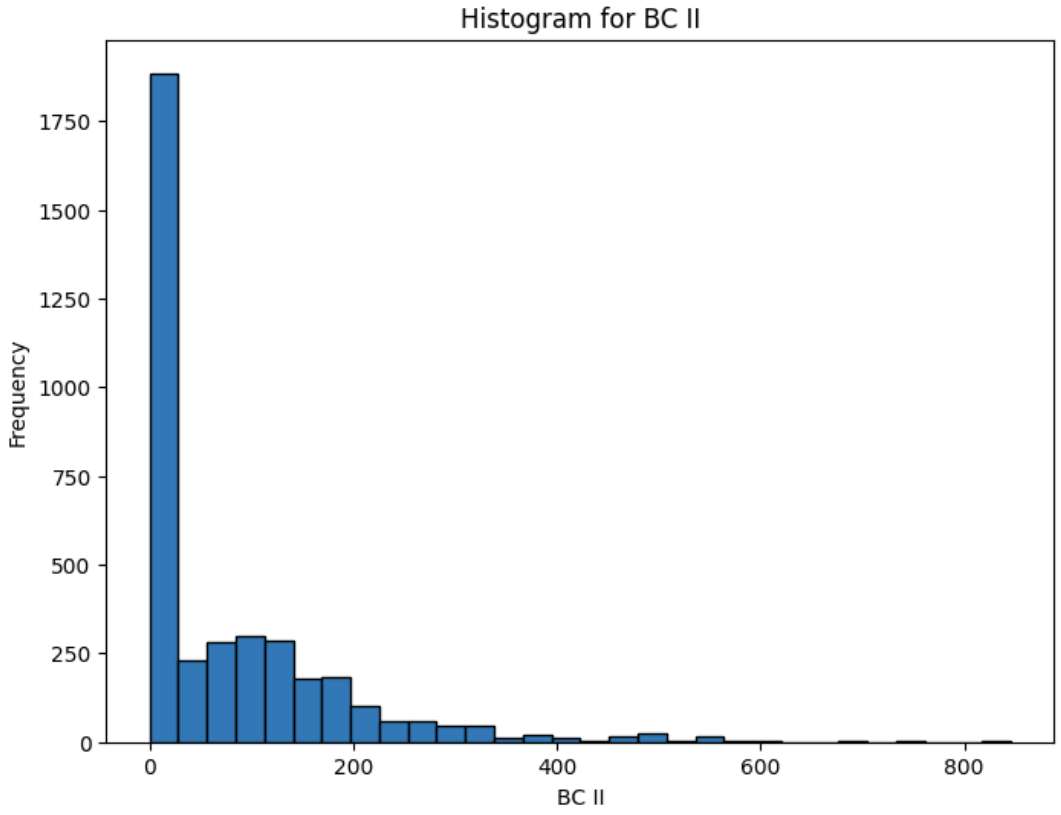
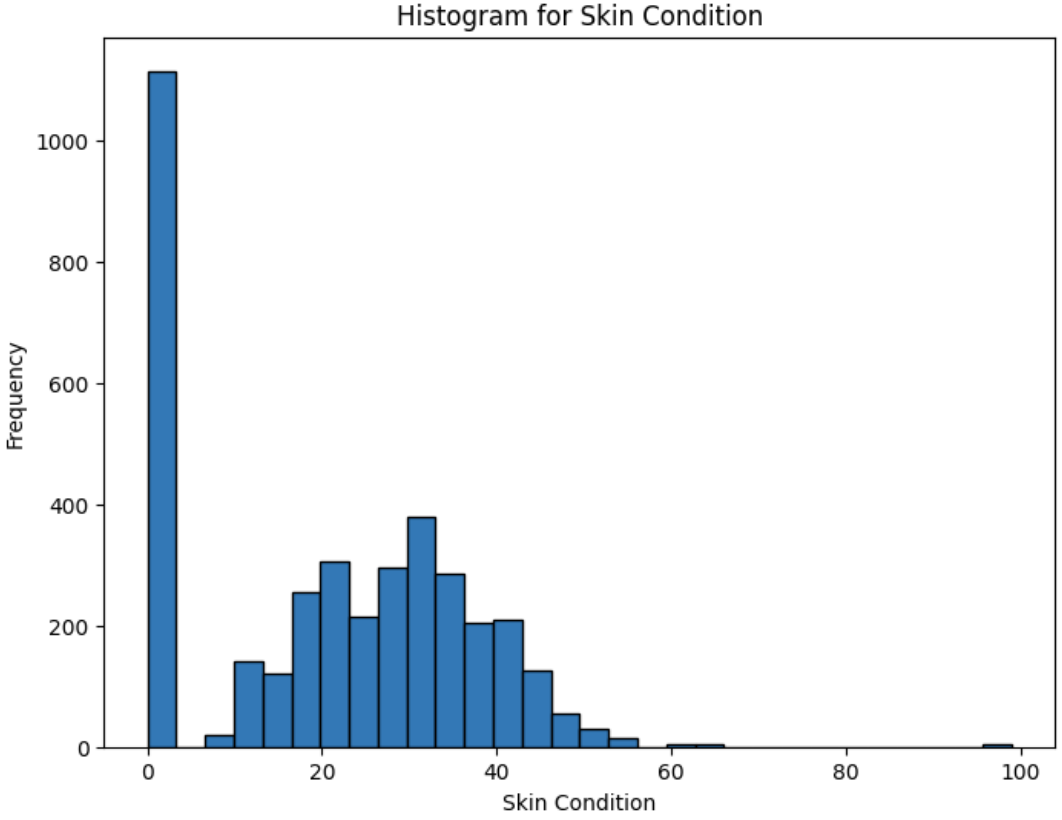
Let's take a look into our initial data:

| | Unique_ID | # Pregnancies | Blood Chemestry~I | Blood Chemisty~II | Blood Chemisty~III | Blood Pressure | Skin Thickness | BMI | Genetic Predisposition Factor | Age | Air Qual'ty Index | $tate | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 5642118.0 | 1.0 | 0.0 | 23.0 | 10.0 | 74.0 | 20.0 | 27.7 | 0.299 | 21.0 | 38.0 | CA | 0.0 |
| 1 | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN |
| 2 | 5642119.0 | 1.0 | 0.0 | 0.0 | 61.0 | 68.0 | 35.0 | 32.0 | 0.389 | 22.0 | 10.0 | CA | 0.0 |
| 3 | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN |
| 4 | 5642120.0 | 1.0 | 0.0 | 0.0 | 172.0 | 48.0 | 20.0 | 24.7 | 0.140 | 22.0 | 77.0 | CA | 0.0 |
| 5 | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN |
| 6 | 5642121.0 | 5.0 | 44.0 | 0.0 | 207.0 | 62.0 | 0.0 | 25.0 | 0.587 | 36.0 | 40.0 | OR | 0.0 |
| 7 | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN |
| 8 | 5642122.0 | 2.0 | 56.0 | 45.0 | 150.0 | 56.0 | 28.0 | 24.2 | 0.332 | 22.0 | 70.0 | CT | 0.0 |
| 9 | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN | NaN |

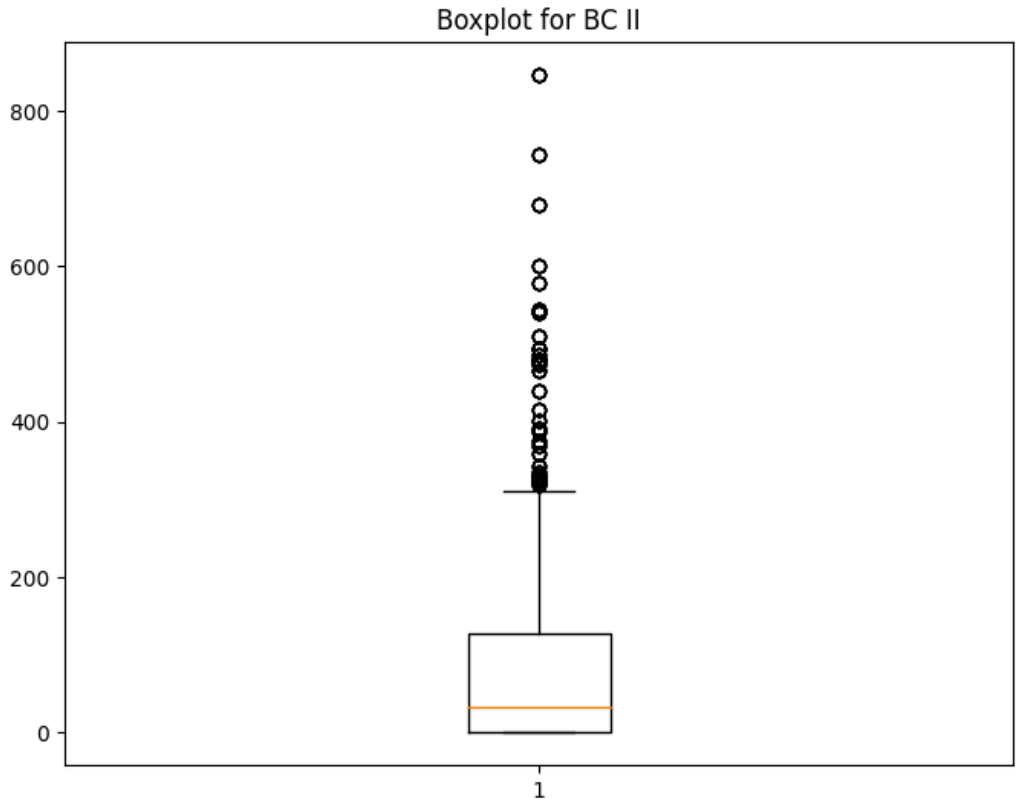
We can see that the data consist a lot of rows with NaN values and names of variables are incorrect comparing with the documentation which we received so we want to correct it and make the data quicker to use.

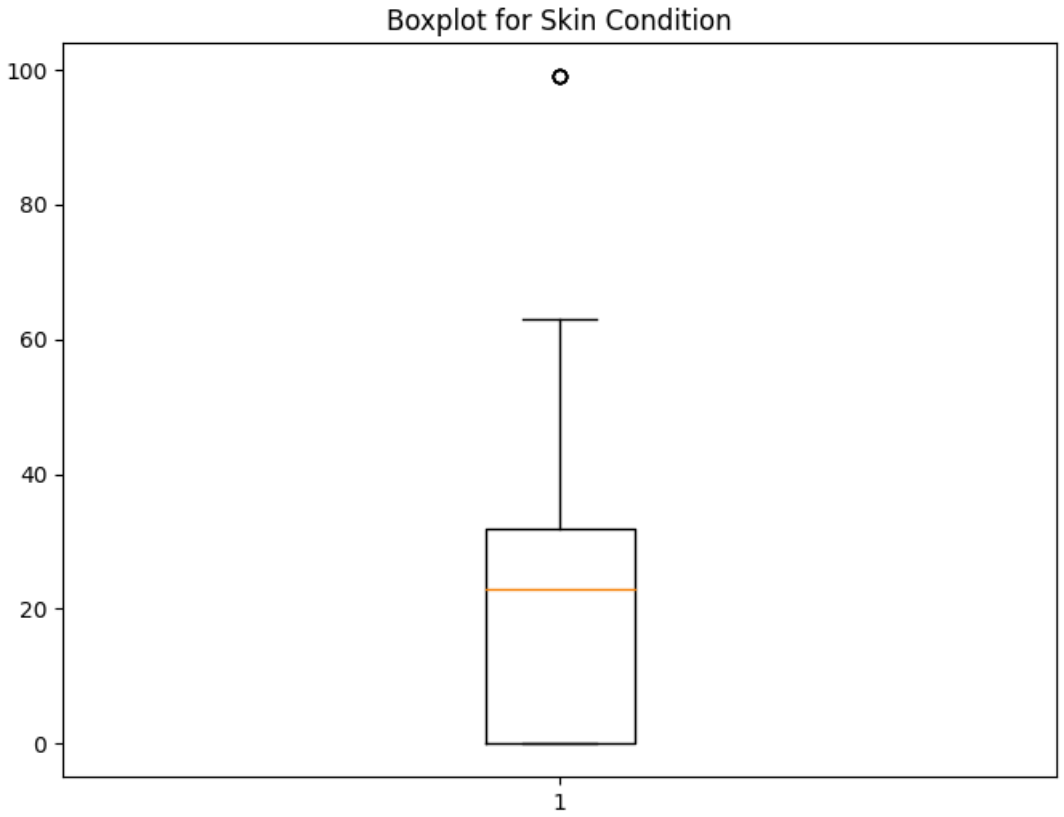
Now let's look deeper into particular variables. Here we present **histograms** for some sample variables...



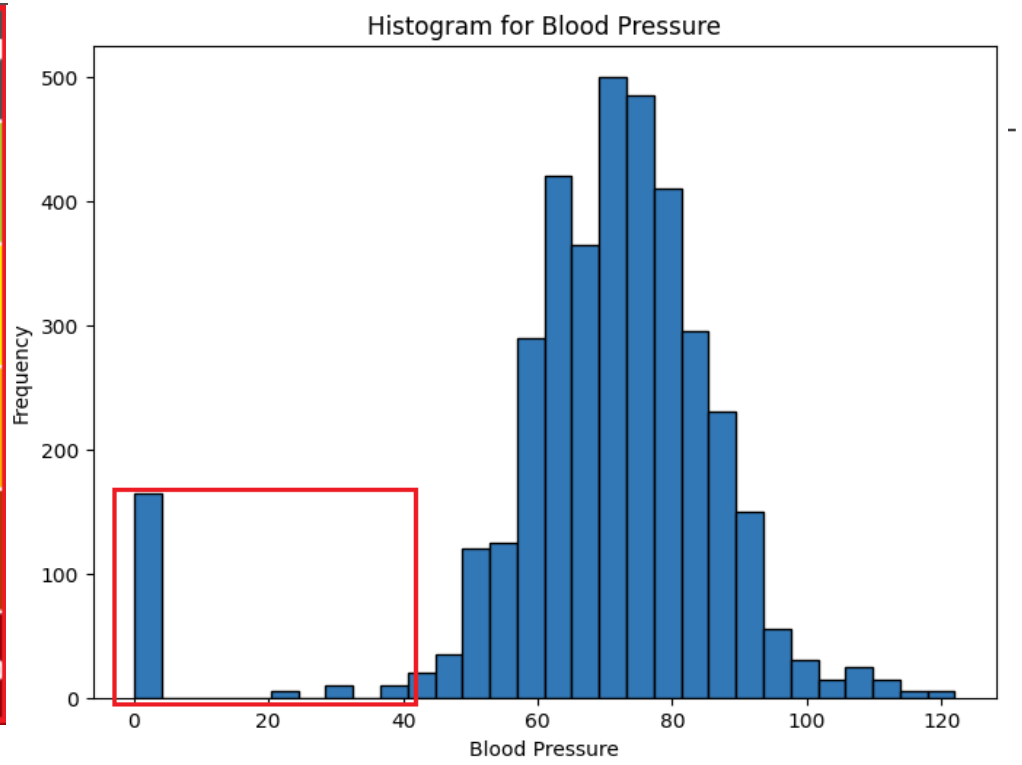
...and **boxplots** for the same sample variables.



Using these graphical representations and leverage well known knowledge we can notice some outlier values. Using commonly known information and information based on histogram we know that blood pressure indicate diastolic pressure and can take the values shown in the diagram below (on the right side):

| BLOOD PRESSURE CATEGORY | SYSTOLIC mm Hg (upper number) | | DIASTOLIC mm Hg (lower number) |
|---|---|---|---|
| NORMAL | LESS THAN 120 | and | LESS THAN 80 |
| ELEVATED | 120 – 129 | and | LESS THAN 80 |
| HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1 | 130 – 139 | or | 80 – 89 |
| HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2 | 140 OR HIGHER | or | 90 OR HIGHER |
| HYPERTENSIVE CRISIS (consult your doctor immediately) | HIGHER THAN 180 | and/or | HIGHER THAN 120 |

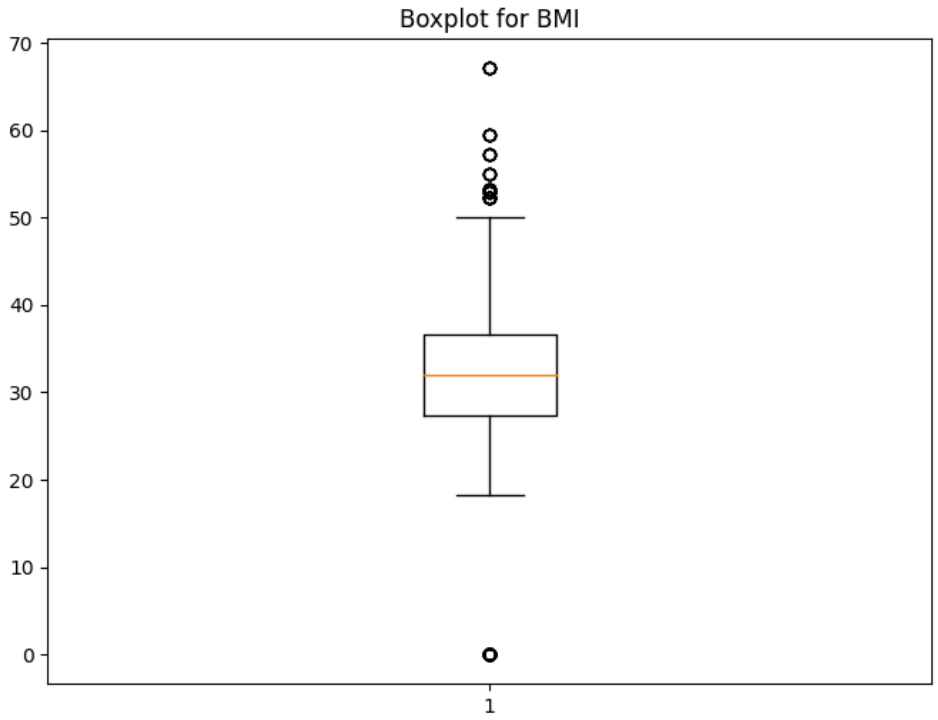



However histogram above shows us that the data consist some outlier values. We decided to set a outlier treshold on 40 or less. For all of outliers we change the value of blood pressure for the mean of blood pressure depending on what outcome record has. If a patient has a disease then we set a blood pressure value as a blood pressure mean for all patients with disease (outcome = 1). Otherwise we set blood pressure value as a blood pressure mean for all patients without disease (outcome = 0).

We decided to make a similar approach for BMI variable. Based on infromation in the table below...

| HEIGHT in | HEIGHT cm | WEIGHT lbs / kgs | 120 / 54.4 | 130 / 59.0 | 140 / 63.5 | 150 / 68.0 | 160 / 72.6 | 170 / 77.1 | 180 / 81.6 | 190 / 86.2 | 200 / 90.7 | 210 / 95.2 | 220 / 99.8 | 230 / 104.3 | 240 / 108.9 | 250 / 113.4 | 260 / 117.9 | 270 / 122.5 | 280 / 127.0 | 290 / 131.5 | 300 / 136.1 | 310 / 140.6 | 320 / 145.1 | 330 / 149.7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4'10" | 124.5 | | 25 | 27 | 29 | 31 | 34 | 36 | 38 | 40 | 42 | 44 | 46 | 48 | 50 | 52 | 54 | 57 | 59 | 61 | 63 | 65 | 67 | 69 |
| 4'11" | 149.9 | | 24 | 26 | 28 | 30 | 32 | 33 | 36 | 38 | 40 | 43 | 45 | 47 | 49 | 51 | 53 | 55 | 57 | 59 | 61 | 63 | 65 | 67 |
| 5'0" | 152.4 | | 23 | 25 | 27 | 29 | 31 | 32 | 35 | 37 | 39 | 41 | 43 | 45 | 47 | 49 | 51 | 53 | 55 | 57 | 59 | 61 | 63 | 65 |
| 5'1" | 154.9 | | 23 | 25 | 26 | 28 | 30 | 32 | 34 | 36 | 38 | 40 | 42 | 44 | 45 | 47 | 49 | 51 | 53 | 55 | 57 | 59 | 61 | 62 |
| 5'2" | 157.5 | | 22 | 24 | 25 | 27 | 29 | 31 | 33 | 35 | 37 | 38 | 40 | 42 | 44 | 46 | 48 | 49 | 51 | 53 | 55 | 57 | 59 | 60 |
| 5'3" | 160.0 | | 21 | 23 | 25 | 27 | 28 | 30 | 32 | 34 | 36 | 37 | 39 | 41 | 43 | 44 | 46 | 48 | 50 | 51 | 53 | 55 | 57 | 59 |
| 5'4" | 162.6 | | 21 | 22 | 24 | 26 | 28 | 29 | 31 | 33 | 34 | 36 | 38 | 40 | 41 | 43 | 45 | 46 | 48 | 50 | 52 | 53 | 55 | 57 |
| 5'5" | 165.1 | | 20 | 22 | 23 | 25 | 27 | 28 | 30 | 32 | 33 | 35 | 37 | 38 | 40 | 42 | 43 | 45 | 47 | 48 | 50 | 52 | 53 | 55 |
| 5'6" | 167.6 | | 19 | 21 | 23 | 24 | 26 | 27 | 29 | 31 | 32 | 34 | 36 | 37 | 39 | 40 | 42 | 44 | 45 | 47 | 49 | 50 | 52 | 53 |
| 5'7" | 170.2 | | 19 | 20 | 22 | 24 | 25 | 27 | 28 | 30 | 31 | 33 | 35 | 36 | 38 | 39 | 41 | 42 | 44 | 46 | 47 | 49 | 50 | 52 |
| 5'8" | 172.7 | | 18 | 20 | 21 | 23 | 24 | 26 | 27 | 29 | 30 | 32 | 34 | 35 | 37 | 38 | 40 | 41 | 43 | 44 | 46 | 47 | 49 | 50 |
| 5'9" | 175.3 | | 18 | 19 | 21 | 22 | 24 | 25 | 27 | 28 | 30 | 31 | 33 | 34 | 36 | 37 | 38 | 40 | 41 | 43 | 44 | 46 | 47 | 49 |
| 5'10" | 177.8 | | 17 | 19 | 20 | 22 | 23 | 24 | 26 | 27 | 29 | 30 | 32 | 33 | 35 | 36 | 37 | 39 | 40 | 42 | 43 | 45 | 46 | 47 |
| 5'11" | 180.3 | | 17 | 18 | 20 | 21 | 22 | 24 | 25 | 27 | 28 | 29 | 31 | 32 | 34 | 35 | 36 | 38 | 39 | 41 | 42 | 43 | 45 | 45 |
| 6'0" | 182.9 | | 16 | 18 | 19 | 20 | 22 | 23 | 24 | 26 | 27 | 29 | 30 | 31 | 33 | 34 | 35 | 37 | 38 | 39 | 41 | 43 | 43 | 45 |
| 6'1" | 185.4 | | 16 | 17 | 19 | 20 | 21 | 22 | 24 | 25 | 26 | 28 | 29 | 30 | 32 | 33 | 34 | 36 | 37 | 38 | 40 | 41 | 42 | 44 |
| 6'2" | 188.0 | | 15 | 17 | 18 | 19 | 21 | 22 | 23 | 24 | 26 | 27 | 28 | 30 | 31 | 32 | 33 | 35 | 36 | 37 | 39 | 40 | 41 | 42 |
| 6'3" | 190.5 | | 15 | 16 | 18 | 19 | 20 | 21 | 23 | 24 | 25 | 26 | 28 | 29 | 30 | 31 | 33 | 34 | 35 | 36 | 38 | 39 | 41 | 41 |
| 6'4" | 193.0 | | 15 | 16 | 17 | 18 | 20 | 21 | 22 | 23 | 24 | 26 | 27 | 28 | 29 | 30 | 32 | 33 | 34 | 35 | 37 | 38 | 39 | 40 |
| 6'5" | 195.6 | | 14 | 15 | 17 | 18 | 19 | 20 | 21 | 23 | 24 | 25 | 26 | 27 | 29 | 30 | 31 | 32 | 33 | 34 | 36 | 37 | 38 | 39 |

Underweight: < 18.5 | Healthy: 18.5 - 24.9 | Overweight: 25 - 29.9 | Obese: 30 - 39.9 | Severely Obese: ≥ 40




...and looking at boxplot we can assume that all values less than 15 and higher than 55 are outliers and we can change it to mean value based on outcome value (as in Blood Pressure).

We notice also some records with outstanding values such as 230-year-old patients.

| | ID | # Pregnancies | BC I | BC II | BC III | Blood Pressure | Skin Condition | BMI | GPF | Age | Air Quality Index | State | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **159** | 5642277 | 1.0 | 97.0 | 82.0 | 262.0 | 64.0 | 19.0 | 18.2 | 0.299 | 230.0 | 58.0 | WI | 0.0 |
| **916** | 5642277 | 1.0 | 97.0 | 82.0 | 158.0 | 64.0 | 19.0 | 18.2 | 0.299 | 230.0 | 58.0 | WI | 0.0 |
| **1673** | 5642277 | 1.0 | 97.0 | 82.0 | 55.0 | 64.0 | 19.0 | 18.2 | 0.299 | 230.0 | 58.0 | WI | 0.0 |
| **2430** | 5642277 | 1.0 | 97.0 | 82.0 | 205.0 | 64.0 | 19.0 | 18.2 | 0.299 | 230.0 | 58.0 | WI | 0.0 |
| **3187** | 5642277 | 1.0 | 97.0 | 82.0 | 307.0 | 64.0 | 19.0 | 18.2 | 0.299 | 230.0 | 58.0 | WI | 0.0 |

We decided to delete them from our dataset due to small volume of it. It was only 5 records so it should not influence on our dataset information.

We noticed that some of the records have 'KU' abbreviation in State column. We assume that those were spelling mistakes and it means Kentucky ('KY'). We excluded 'ID' column from our dataset – it does not gives us any valuable information.

| | ID | # Pregnancies | BC I | BC II | BC III | Blood Pressure | Skin Condition | BMI | GPF | Age | Air Quality Index | State | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **12** | 5642130 | 2.0 | 68.0 | 66.0 | 49.0 | 70.0 | 32.0 | 25.0 | 0.187 | 25.0 | 0.0 | KY | 0.0 |
| **45** | 5642163 | 2.0 | 81.0 | 76.0 | 67.0 | 72.0 | 15.0 | 30.1 | 0.547 | 25.0 | 2.0 | KY | 0.0 |
| **101** | 5642219 | 1.0 | 90.0 | 59.0 | 89.0 | 62.0 | 18.0 | 25.1 | 1.268 | 25.0 | 90.0 | KY | 0.0 |
| **116** | 5642234 | 1.0 | 92.0 | 41.0 | 14.0 | 62.0 | 25.0 | 19.5 | 0.482 | 25.0 | 36.0 | KY | 0.0 |
| **139** | 5642257 | 1.0 | 95.0 | 38.0 | 53.0 | 66.0 | 13.0 | 19.6 | 0.334 | 25.0 | 52.0 | KY | 0.0 |
| **214** | 5642333 | 6.0 | 103.0 | 190.0 | 269.0 | 72.0 | 32.0 | 37.7 | 0.324 | 55.0 | 31.0 | KY | 0.0 |
| **288** | 5642407 | 1.0 | 112.0 | 176.0 | 316.0 | 72.0 | 30.0 | 34.4 | 0.528 | 25.0 | 60.0 | KY | 0.0 |
| **311** | 5642431 | 1.0 | 116.0 | 180.0 | 257.0 | 78.0 | 29.0 | 36.1 | 0.496 | 25.0 | 60.0 | KY | 0.0 |
| **512** | 5642638 | 10.0 | 101.0 | 0.0 | 60.0 | 86.0 | 37.0 | 45.6 | 1.136 | 38.0 | 78.0 | KU | 1.0 |
| **531** | 5642657 | 5.0 | 109.0 | 129.0 | 318.0 | 62.0 | 41.0 | 35.8 | 0.514 | 25.0 | 92.0 | KY | 1.0 |
| **542** | 5642668 | 9.0 | 112.0 | 0.0 | 22.0 | 82.0 | 24.0 | 28.2 | 1.282 | 50.0 | 24.0 | KY | 1.0 |
| **623** | 5642749 | 5.0 | 139.0 | 160.0 | 56.0 | 80.0 | 35.0 | 31.6 | 0.361 | 25.0 | 89.0 | KU | 1.0 |
| **656** | 5642782 | 12.0 | 151.0 | 271.0 | 15.0 | 70.0 | 40.0 | 41.8 | 0.742 | 38.0 | 32.0 | KY | 1.0 |

Now we can describe some descriptive statistics to our variables...

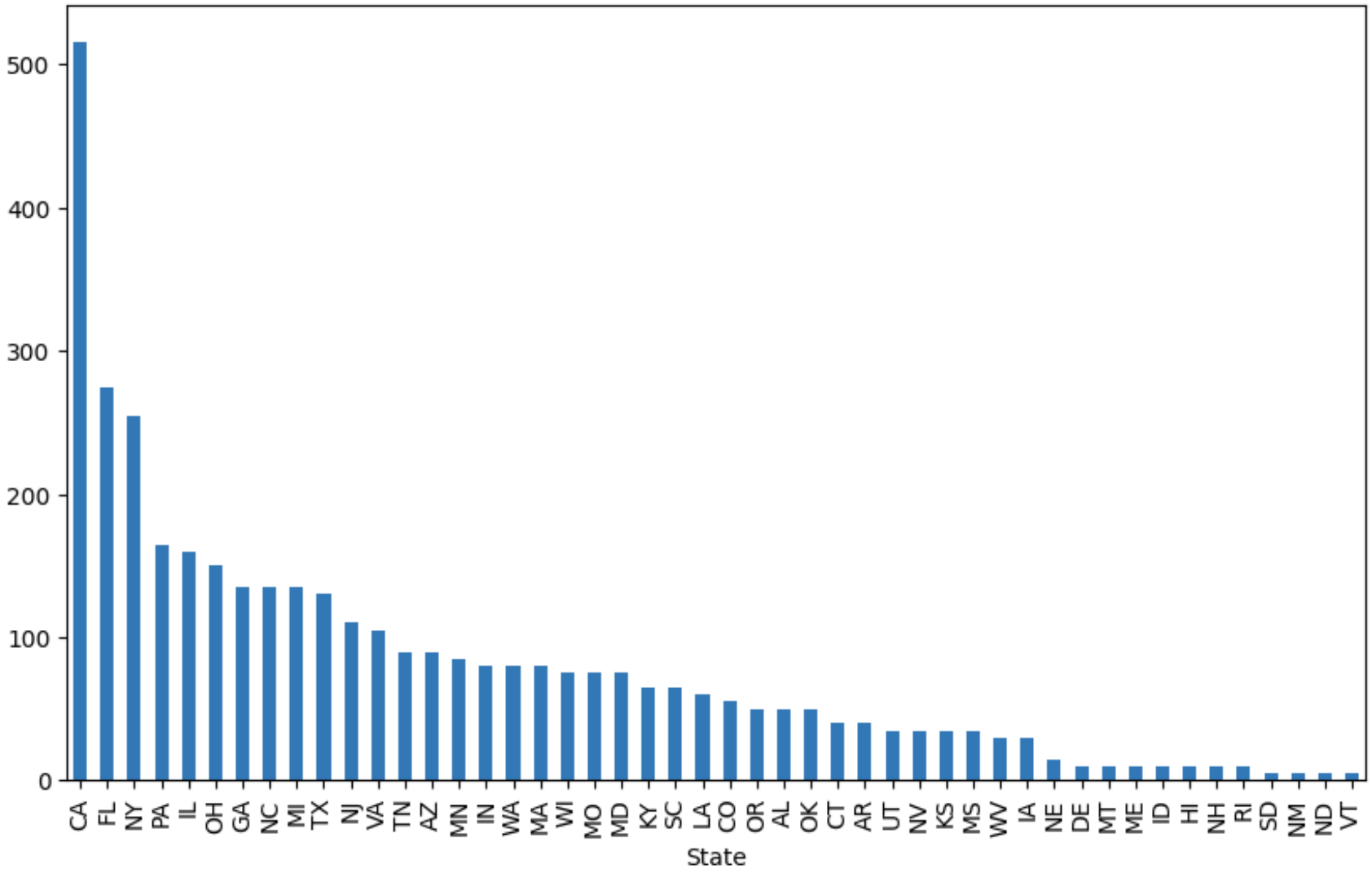
| | # Pregnancies | BC I | BC II | BC III | Blood Pressure | Skin Condition | BMI | GPF | Age | Air Quality Index | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 |
| **mean** | 3.863757 | 120.850529 | 80.101852 | 174.383862 | 72.345439 | 20.646825 | 32.379282 | 0.472257 | 33.297619 | 49.072751 | 0.350529 |
| **std** | 3.354539 | 32.031262 | 115.727797 | 100.917430 | 12.057181 | 15.984856 | 6.619298 | 0.332240 | 11.768587 | 29.169210 | 0.477199 |
| **min** | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 30.000000 | 0.000000 | 18.200000 | 0.078000 | 21.000000 | 0.000000 | 0.000000 |
| **25%** | 1.000000 | 99.000000 | 0.000000 | 88.000000 | 64.000000 | 0.000000 | 27.600000 | 0.242500 | 24.000000 | 24.000000 | 0.000000 |
| **50%** | 3.000000 | 117.000000 | 30.500000 | 175.000000 | 72.000000 | 23.000000 | 32.050000 | 0.372500 | 29.000000 | 49.000000 | 0.000000 |
| **75%** | 6.000000 | 141.000000 | 127.250000 | 259.000000 | 80.000000 | 32.000000 | 36.500000 | 0.626250 | 41.000000 | 72.250000 | 1.000000 |
| **max** | 17.000000 | 199.000000 | 846.000000 | 350.000000 | 122.000000 | 99.000000 | 55.000000 | 2.420000 | 81.000000 | 100.000000 | 1.000000 |

… and the same descriptive statistics after normalizing the data which we apply to avoid big influence of potential outliers which we did not manage to mitigate and have a possibility to compare variables to each other.

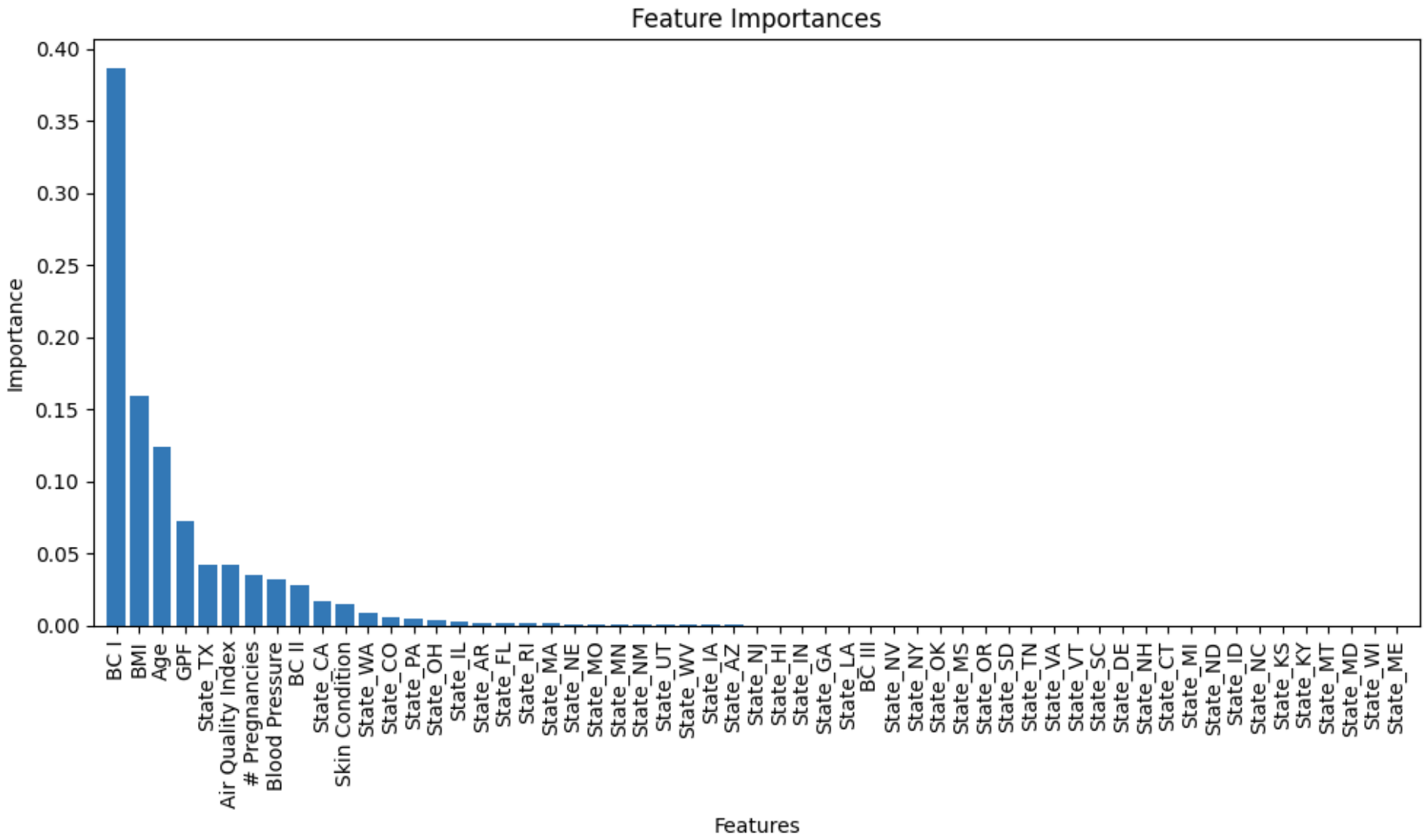
| | # Pregnancies | BC I | BC II | BC III | Blood Pressure | Skin Condition | BMI | GPF | Age | Air Quality Index | Outcome |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 | 3780.000000 |
| **mean** | 0.227280 | 0.607289 | 0.094683 | 0.498240 | 0.460277 | 0.208554 | 0.385307 | 0.168342 | 0.204960 | 0.490728 | 0.350529 |
| **std** | 0.197326 | 0.160961 | 0.136794 | 0.288336 | 0.131056 | 0.161463 | 0.179872 | 0.141862 | 0.196143 | 0.291692 | 0.477199 |
| **min** | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| **25%** | 0.058824 | 0.497487 | 0.000000 | 0.251429 | 0.369565 | 0.000000 | 0.255435 | 0.070239 | 0.050000 | 0.240000 | 0.000000 |
| **50%** | 0.176471 | 0.587940 | 0.036052 | 0.500000 | 0.456522 | 0.232323 | 0.376359 | 0.125747 | 0.133333 | 0.490000 | 0.000000 |
| **75%** | 0.352941 | 0.708543 | 0.150414 | 0.740000 | 0.543478 | 0.323232 | 0.497283 | 0.234095 | 0.333333 | 0.722500 | 1.000000 |
| **max** | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |

We noticed that the blood chemical composition type III and the air quality index had the largest deviations from the mean, as indicated by the high value of the standard deviation. It means that they are the most diverse. On the other hand, Blood Chemistry type II and Genetic Prediposition Factor are the variables most clustered to the mean, which means less variation in their measurements.

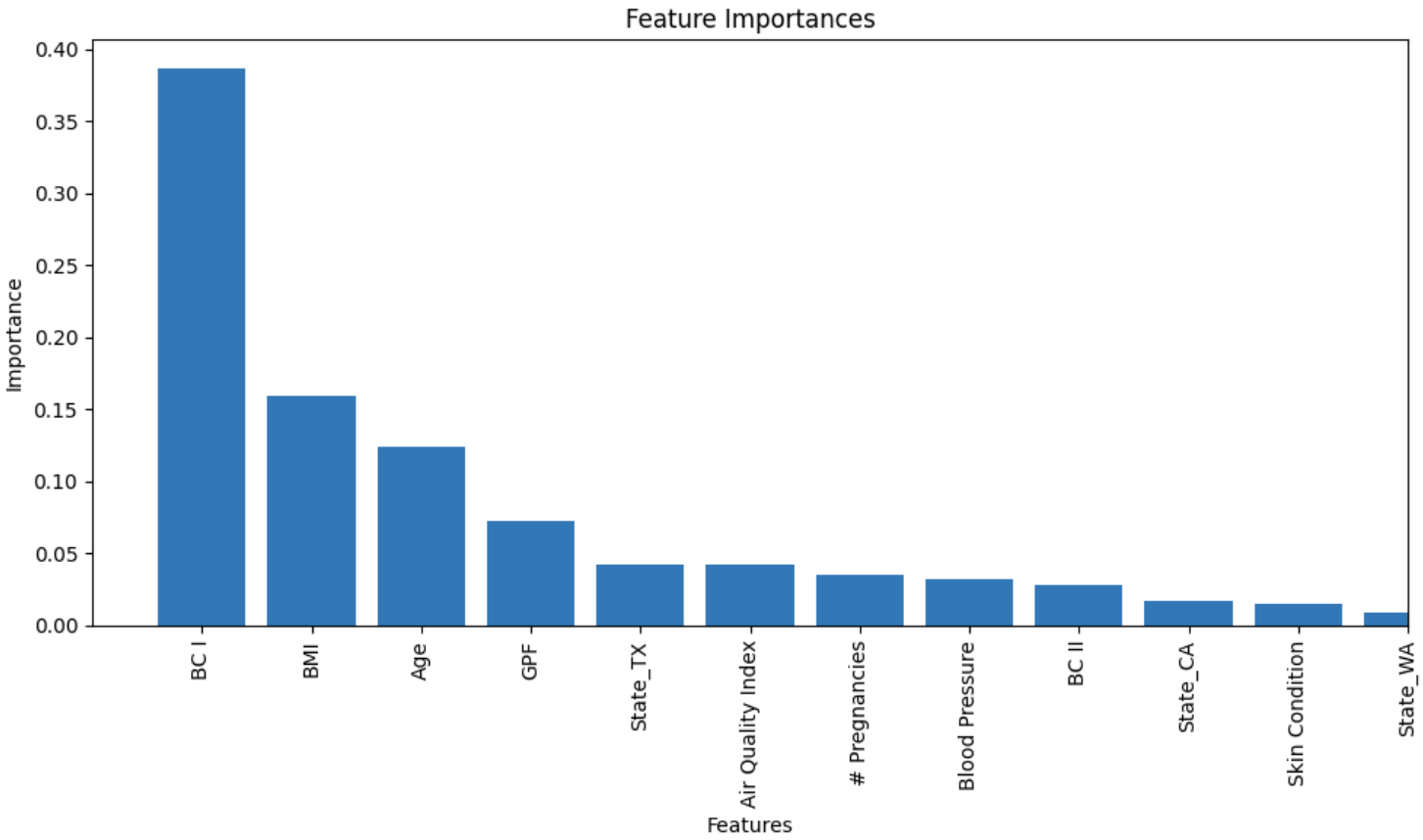
The distribution of data in terms of residence status is also valuable information. We can see it in the chart below.



Let's look at some patterns which we notice in the data. The data consist a lot of variables so we want to check which of them have the greatest impact on the outcome of whether a given patient is sick or not.
Here we can see a feature importance ranking...

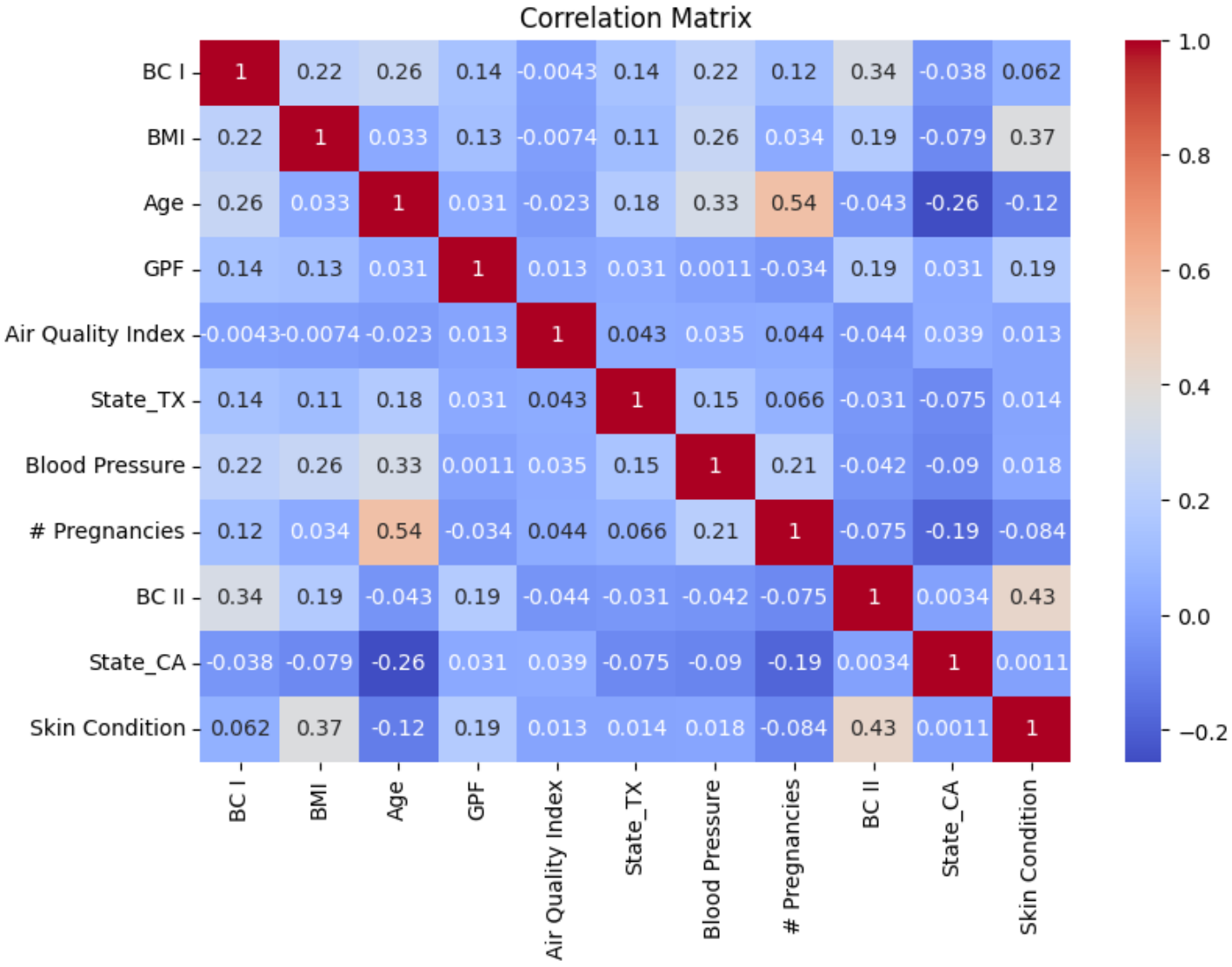


We focus on top 10 most impactful features.




As we can see Blood Chemistry type I, BMI and Age are the most impactful variables on the result of the outcome. What is also interesting is that living in the states of TX or CA was in the top 10 most influential factors for the presence of the disease.

Let's look also at some correlations between variables which we can see in the data. We use only top 10 most influence features to make our chart more transparent and easy to read.



Here is some results that we can notice:

- the largest positive correlation is between age and the number of pregnancies (0.54)

- skin condition has high positive correlation rates between Blood Chemistry II and BMI (0.43 and 0.37, respectively)

- Blood Chemistry I and Blood Chemistry II are also related to each other with a high positive correlation (0.34)

- positive correlation index between age and blood pressure (0.33)

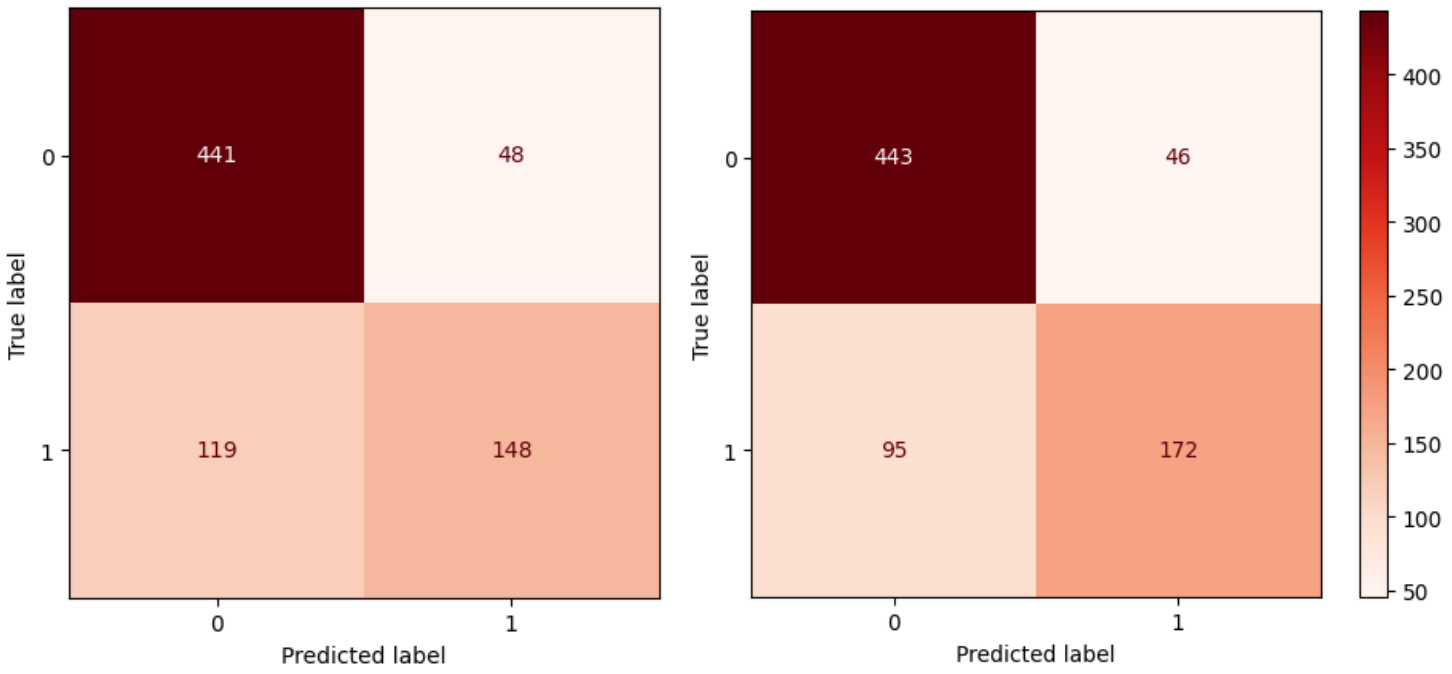
- negative correlation coefficient between age and residence in the state of CA (-0.26)

We can interpret them as follows:
- older patients have more children
- poor skin condition may indicate a high BMI and poor blood test results
- a high Blood Chemistry type I rate may also indicate high Blood Chemistry type II rates
- with age, patients' blood pressure worsens
- patients residing in California are young people

We can confirm that California patients are young based on the average age in our database. We can also see what the average age is in the state of Texas. What's more interesting, investigating why the high importance value of residence in the state of Texas is, we can notice that out of 130 records from this state, 125 of them have a disease. This may be important information when ordering specific tests for people from the state of Texas.

```
Average age in CA: 25.737864077669904
Average age in other states: 34.49004594180705
Average age in TX : 44.80769230769231
Average age in other states: 32.88767123287671
Number of records from TX: 130
Number of records from TX with disease: 125.0
```

We have also created 2 predictive models that will help predict whether a given patient with specific variables is sick or not. The graphs below show the models' accuracy to be 78% and 81.3%, respectively.



# Recommendations

- regular screening for Blood Chemistry type I and BMI,
- more frequent tests for older patients,
- due to the higher Genetic Predisposition Factor feature importance, monitoring family history related to the disease,
- detailed and more comprehensive tests for Texas patients,
- paying attention to the condition of patients' skin,
- more frequent blood pressure monitoring in older people,
-more attention when recording patient data.

# References:


- https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings
- https://www.builtlean.com/bmi-chart/